ASSIGNMENT:8.3

COURSE: AI Assisted Coding

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BATCH NO : 09

Lab 8:

Test-Driven Development with AI – Generating and Working with Test Cases

Lab Objectives
• Introduce TDD using AI
• Generate test cases before implementation
• Emphasize testing and validation
• Encourage clean, reliable code
Lab Outcomes
Students will be able to:
• Write AI-generated test cases
Week4 -
Wednesday
• Implement code using test-first approach
• Validate using unittest
• Analyze test coverage
• Compare AI vs manual tests

Task 1:Email Validation

Prompt:

Generate unittest test cases for a function `is_valid_email(email)` that validates email addresses.
Conditions:

- Must contain `@` and `.`
- Must not start or end with special characters
- Only one `@` allowed
- Include valid and invalid examples

Code:

```
import re

def is_valid_email(email):
    if not isinstance(email, str):
        return False

    if email.count('@') != 1:
        return False

    if '@' not in email or '.' not in email:
        return False

    if email[0] in ".@_" or email[-1] in ".@_":
        return False

    pattern = r'^[A-Za-z0-9][A-Za-z0-9._]*@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'
    return bool(re.match(pattern, email))
```

```
# Example
print(is_valid_email("test@example.com"))    # True
print(is_valid_email("test@@example.com"))   # False
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   Python

PS C:\Users\V.NITHIN\.vscode\python coding> & C:/Users/V.NITHIN/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/V.NITHIN/.vscode/p
ython coding/import unittest.py"
True
False
PS C:\Users\V.NITHIN\.vscode\python coding>
```

Task 2: Grade Assignment

Prompt:

Generate unittest test cases for `assign_grade(score)`.
Grading rules:

- 90–100 → A
- 80–89 → B
- 70–79 → C
- 60–69 → D
- Below 60 → F
  Include boundary values and invalid inputs.

Code:

```
def assign_grade(score):
    if not isinstance(score, (int, float)):
        return "Invalid Input"

    if score < 0 or score > 100:
```

```python
        return "Invalid Input"

    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    elif score >= 60:
        return "D"
    else:
        return "F"


# Example
print(assign_grade(95))    # A
print(assign_grade(75))    # C
print(assign_grade(-5))    # Invalid Input
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   Python

PS C:\Users\V.NITHIN\.vscode\python coding> & C:/Users/V.NITHIN/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/V.NITHIN/.vscode/p
ython coding/import unittest.py"
A
C
Invalid Input
PS C:\Users\V.NITHIN\.vscode\python coding>
```

Task 3: Sentence Palindrome Checker

Prompt:

Generate unittest test cases for `is_sentence_palindrome(sentence)`.

- Ignore case, spaces, and punctuation
- Include both palindrome and non-palindrome sentences
- Example: "A man a plan a canal Panama"

Code:

```python
import re

def is_sentence_palindrome(sentence):
    if not isinstance(sentence, str):
        return False

    cleaned = re.sub(r'[^A-Za-z0-9]', '', sentence).lower()
    return cleaned == cleaned[::-1]


# Example
print(is_sentence_palindrome("A man a plan a canal Panama"))
print(is_sentence_palindrome("Hello World"))
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

PS C:\Users\V.NITHIN\.vscode\python coding> & C:/Users/V.NITHIN/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/V.NITHIN/.vscode/p
ython coding/import unittest.py"
True
False
PS C:\Users\V.NITHIN\.vscode\python coding>
```

Task 4: ShoppingCart Class

Prompt:

Generate unittest test cases for a `ShoppingCart` class with methods:

- `add_item(name, price)`
- `remove_item(name)`
- `total_cost()`

  Test addition, removal, total calculation, and empty cart scenario.

code:

```
class ShoppingCart:
    def __init__(self):
        self.items = {}

    def add_item(self, name, price):
        if price >= 0:
            self.items[name] = price

    def remove_item(self, name):
        if name in self.items:
            del self.items[name]

    def total_cost(self):
        return sum(self.items.values())


# Example
cart = ShoppingCart()
cart.add_item("Book", 500)
cart.add_item("Pen", 50)
print(cart.total_cost())    # 550

cart.remove_item("Pen")
print(cart.total_cost())    # 500
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python

PS C:\Users\V.NITHIN\.vscode\python coding> & C:/Users/V.NITHIN/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/V.NITHIN/.vscode/p
ython coding/import unittest.py"
550
500
PS C:\Users\V.NITHIN\.vscode\python coding>
```

Task 5: Date Format Conversion

Prompt:

Generate unittest test cases for `convert_date_format(date_str)`.

- Input format: YYYY-MM-DD
- Output format: DD-MM-YYYY
- Include invalid date formats.

code:

```python
def convert_date_format(date_str):
    if not isinstance(date_str, str):
        return "Invalid Format"

    parts = date_str.split('-')
    if len(parts) != 3:
        return "Invalid Format"

    year, month, day = parts
    if len(year) != 4 or len(month) != 2 or len(day) != 2:
        return "Invalid Format"

    return f"{day}-{month}-{year}"


# Example
print(convert_date_format("2023-10-15"))
print(convert_date_format("15-10-2023"))
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python

PS C:\Users\V.NITHIN\.vscode\python coding> & C:/Users/V.NITHIN/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/V.NITHIN/.vscode/p
ython coding/import unittest.py"
15-10-2023
Invalid Format
PS C:\Users\V.NITHIN\.vscode\python coding>
```